JUMP AROUND

2D Platformer that slowly teaches certain platforming techniques over the course of the game and challenges the player in unique and interesting ways through various technique combinations.

1. Project Info

Game Title: JUMP AROUND
Author: Nathan Cho
Platform: PC
Engine/Tools: Unity

2. Game Overview & Vision

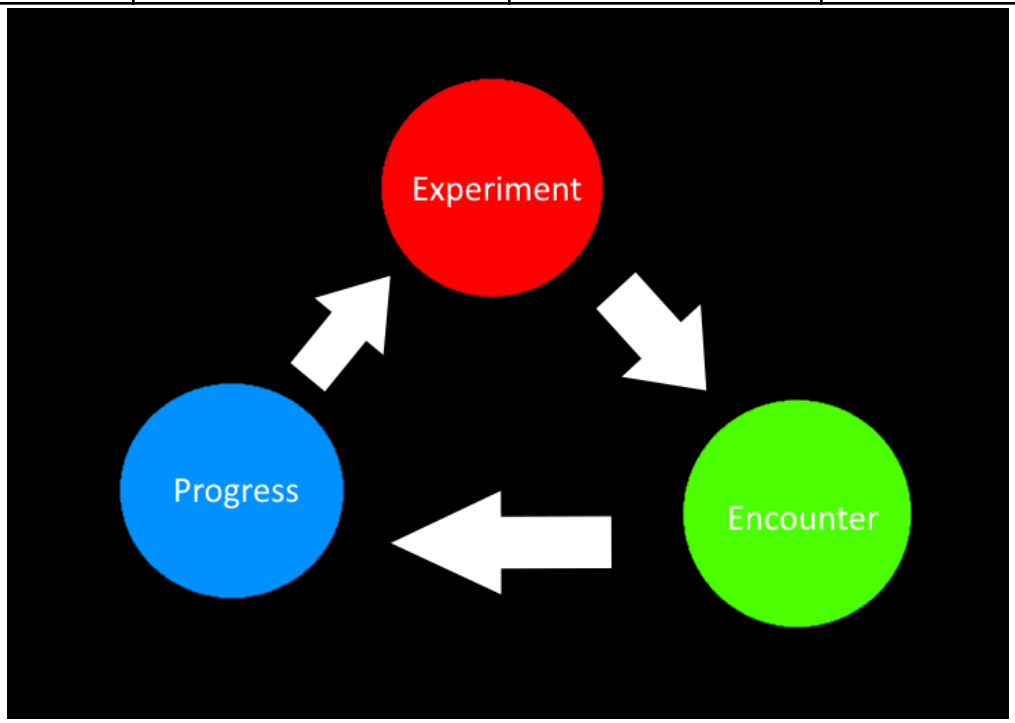
High Concept: A 2D platformer focused on progressive learning of mechanics through bite-sized levels. The player is mainly challenged by their own knowledge of these mechanics and how they interact with each passing level.
Game Summary: Series of separate levels with a set start and end point that challenges players to engage and utilize certain interactions/mechanics that are embedded into each level map. As the player progresses, the challenge increases.
Player Journey Snapshot: In the first 10 minutes, the player will be engaged in only the basics of moving their player character completing typical
Unique Selling Points:
Target Audience
Genre & Inspirations: Metroidvania
Tone & Look & Feel: Pixelart

3. Core Gameplay Design

Core Gameplay Loop: Player enters levels with a clear endpoint shown, must figure out how to leverage the map elements to navigate to the end of the level.

Step	Action	Change	Reward
1 - Experiment	Experiment with physics-based obstacles	Gain real experience controlling the player avatar	Sense of autonomy and control in a new environment
2 - Encounter	Use previously learned techniques to navigate hostile threats	Introduces challenge for the player to use their newfound knowledge against	Players get the challenge and accomplishment after applying what they've learned to defeat new enemies

3 - Progress	Completing a level	Sections the game into manageable chunks which contain their own unique challenges	Accomplishment via completion of a section, representing mastery of the skills used in the individual level



<u>Player Progression:</u> Player progression is strongly tied to knowledge gained by interacting and experimenting with map elements
<u>Controls & Input:</u> WASD and the space bar and the main movement and jump controls

Input	Action	Notes
A / D	Move left / right	
Space / W	Jump	Hold = full, tap = short hop
Left Mouse	Shoot arrow	Shoots in the direction the player is facing
Escape	Pause Menu	

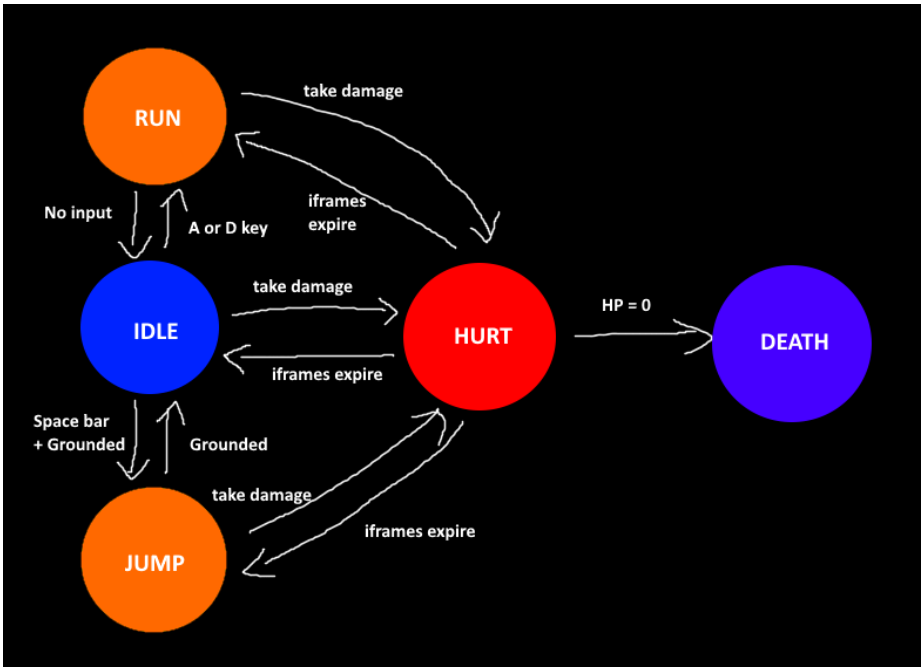
<u>Difficulty & Challenge:</u> Difficulty & challenge increase as players go through sequential levels. With mechanics established in a level being combined or enhanced with other mechanics to provide additional challenge

Feedback Systems: There will be audio queues to indicate the player character state such as running, jumping, or colliding into an object.

Event	Visual Feedback	Audio Feedback	Physical Feedback
Player jumps		Jump SFX	
Player lands			Camera shake
Player hit	Flash + invincibility frames	Hit SFX	Camera shake
Arrow hits enemy		Hit SFX	
Enemy dies	Death animation		
Coin collect		Coin SFX	
Player dies	Death animation		Camera shake

4. Game Mechanics & Systems

Core Mechanics: The player is able to jump and run in order to platform through the level
Side Mechanics: None
Interaction Mechanics: There will be certain elements that the player can interact with by either colliding or manually interacting with that are embedded with the level.
Player State Machine:



Economy & Resources:

Resource	How Earned	How Spent	Notes
Arrows	N/A	Shoots from the player. On contact with enemies, enemies take damage	Infinite ammo, limited fire rate
Hearts	Start with 3	Every time the player is hit, a heart is spent. When hearts is set to 0, the player loses and the game is over	N/A

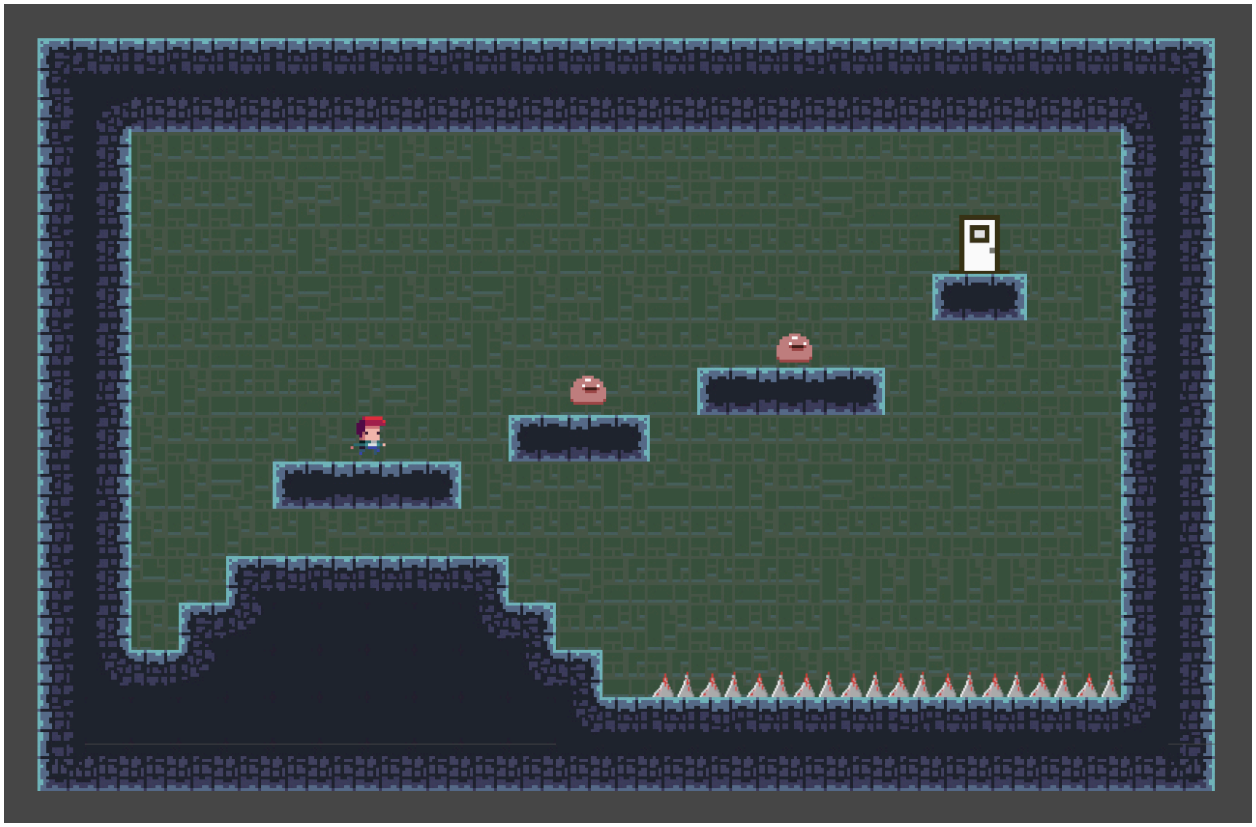
5. Narrative, World & Characters

Themes & Emotional Arc: None
Story Overview: None
Worldbuilding: None
Main Characters: None
Player Agency: None

6. Level & Encounter Design

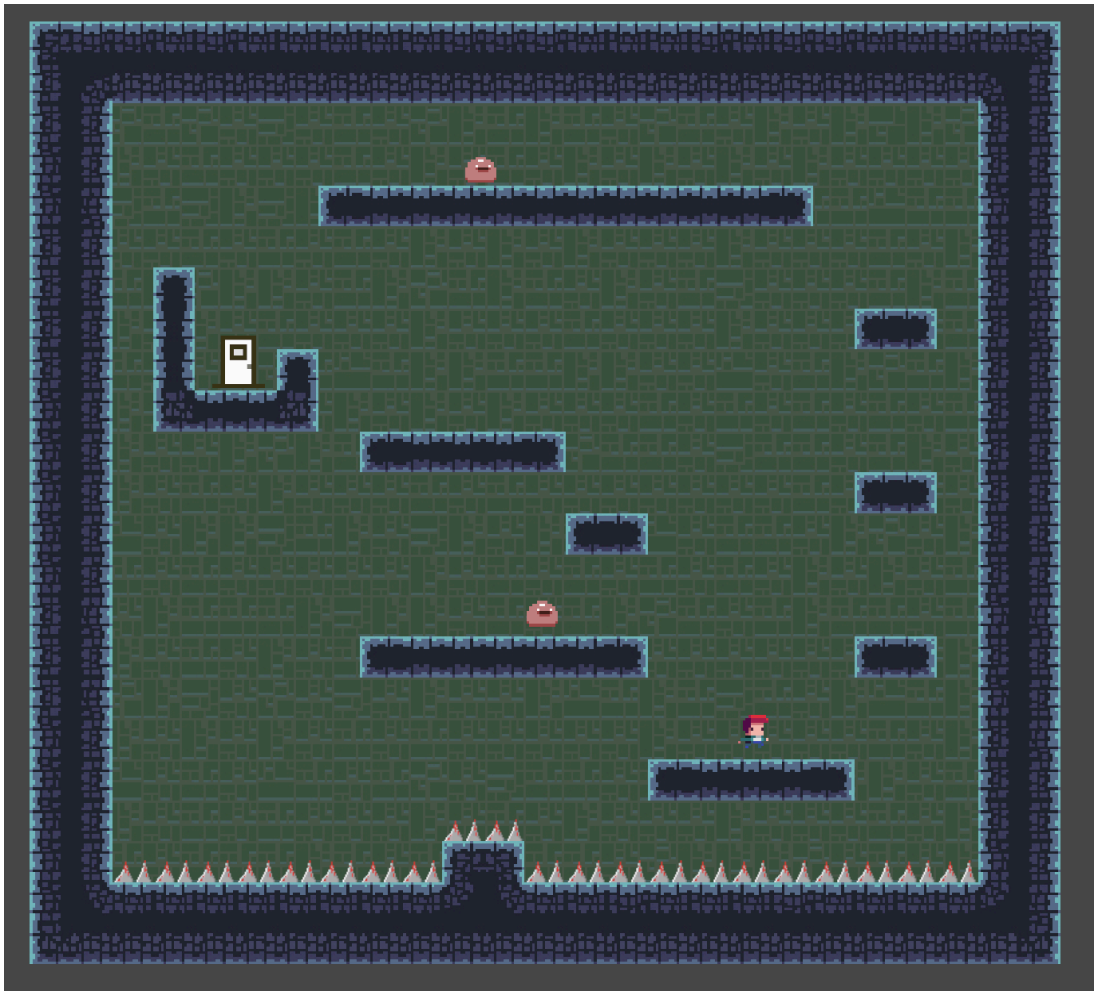
Objectives: Navigate your way through a platforming puzzle to reach a designated endpoint
Structure & Layout: Each level is atomic and sequential meaning there is no carry over between levels and they are encountered in a set order.

Level 1 Overview



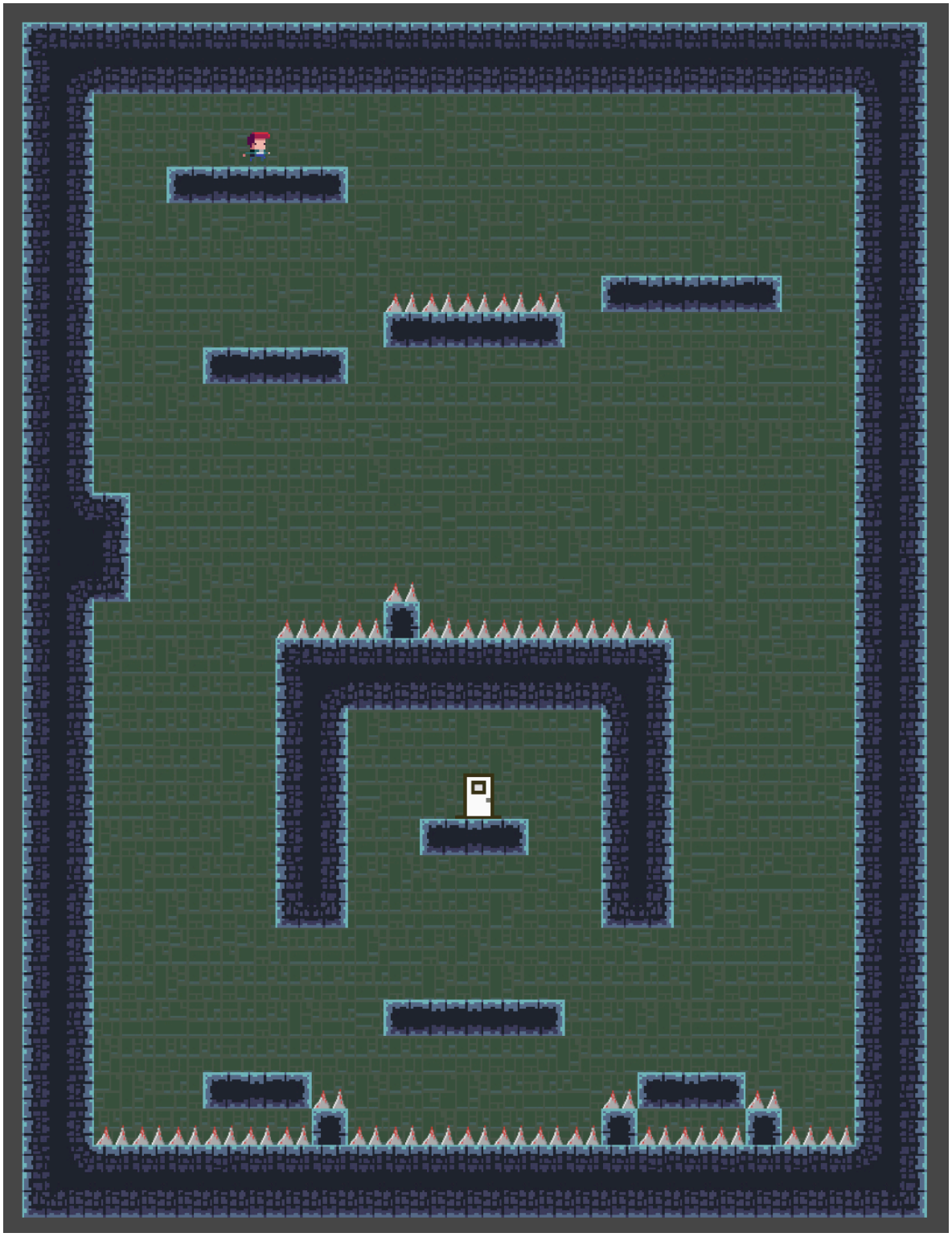
This level introduces all of the basic elements of the game. How enemies interact with the player, the dangers of hazards, and the platforming mechanics necessary to navigate the level.

Level 2 Overview



This level is meant to offer more of a platforming challenge and force the player to a route that tests mid-air maneuverability while avoiding or fighting enemies along the path

Level 3 Overview



This level introduces even more verticality and engages the falling mechanic where the player must navigate this level from top-to-bottom rather than bottom-to-top

Gameplay Flow: The player continuously tries to navigate through the level. By completing a level, they are immediately spawned into the next level.
Interactive Elements: There will be jump pads, levers which manipulate the environment, and items which require player collision to activate.
Progression Gates: Progression is gated by the player's ability to complete a given level. If they can't, there is no progression
Environmental Storytelling: None

7. **UI, UX, & Accessibility**

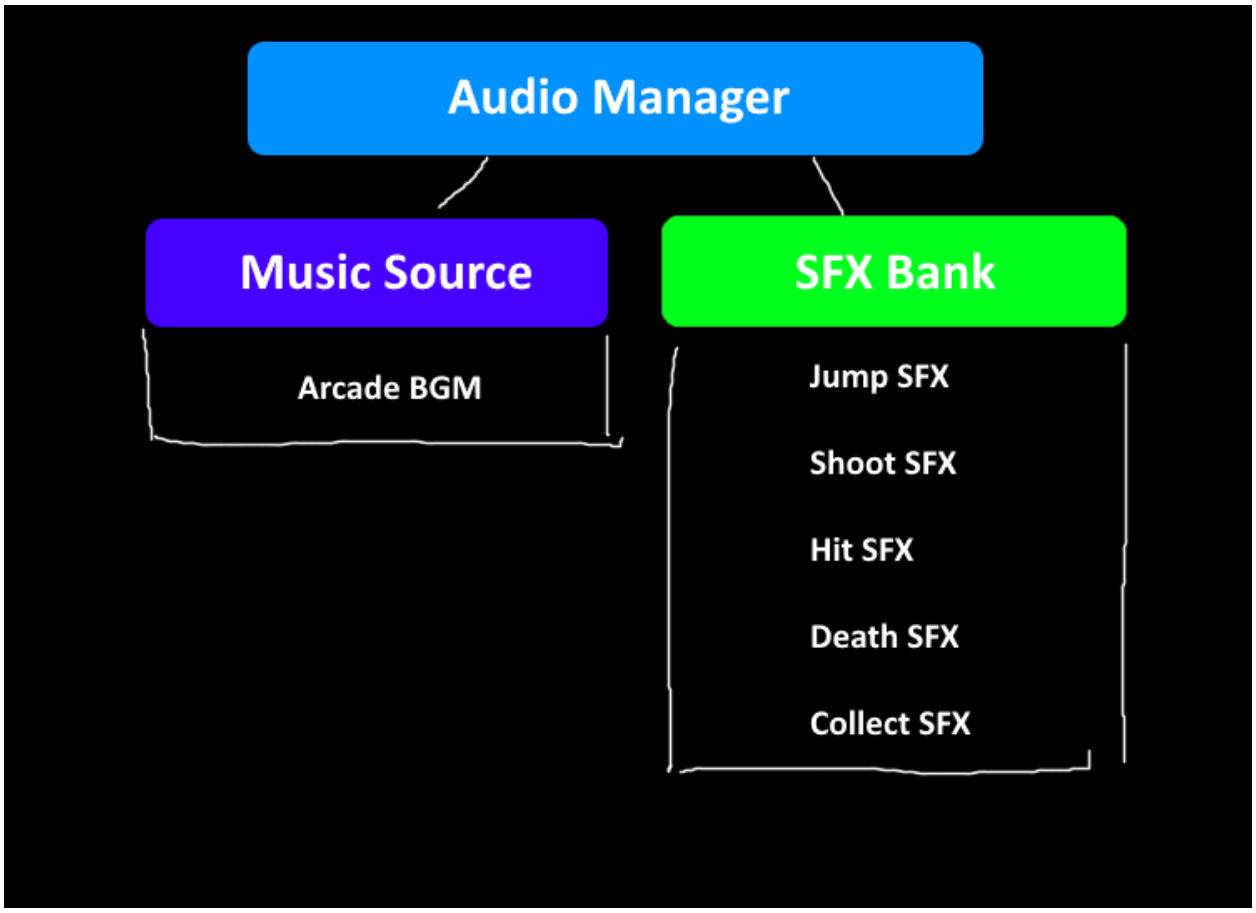
HUD Layout

Core UI Elements: A health bar showing the remaining amount of hits the player can take before being reset to the start of the level

Element	Position	Description
Hearts	Top Left	3 heart icons
Lives Counter	Top Left	Decrements when all 3 hearts are lost

Menus & Screens: There will be a pause menu and a level select menu where a player can replay any of the unlocked levels
HUD Design: The health bar will be shown in the top left at all times
UX Flow: From the main menu, when the player clicks Play, they will be spawned at the start location of the latest level that they unlocked

Onboarding: The game will be fairly hands off for teaching besides the basic movements, allowing the user to experiment with the environment and gain a deeper understanding of the mechanics by themselves.
Accessibility Options: None

8. **Audio Design**

FMOD Event Hierarchy:
Unfortunately, I was not able to participate in the FMOD exploration so I did not use FMOD. However there is an SFX bank using sounds posted on freesounds.org